

27-07-2025

Agenda - start - 1

- (1) $P(A) * P(B)$ AND $\rightarrow *$
OR $\rightarrow +$
 - (2) conditional probability
 - (3) probability question
 - (4) Bayes theorem
 - (5) PMF, PDF & CDF
-

p.1 Draw 2 cards (with replacement)



$\rightarrow$ 1 card (51)



$\rightarrow$ 1 card

first ace $\rightarrow$

$f = 4$

$n = 52$

$$P(\text{first ace}) = \frac{4}{52} = \frac{1}{13}$$

event: Both Aces



$\rightarrow$

52



$\rightarrow$ 4 Ace

1 card

$\rightarrow$

$$\frac{4}{52} = \frac{1}{13}$$

$$P(E) = \frac{\text{favorable}}{\text{total}}$$

$$\text{first_card} = \frac{1}{13}$$

$$\text{second_card} = \frac{4}{52} = \frac{1}{13}$$

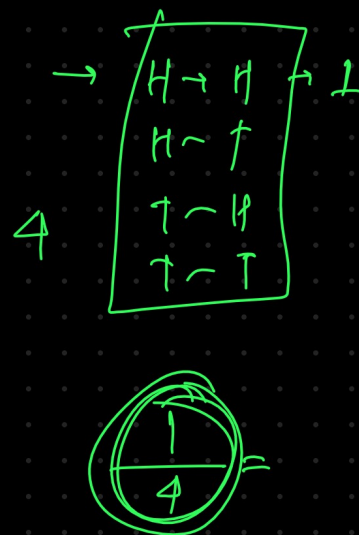
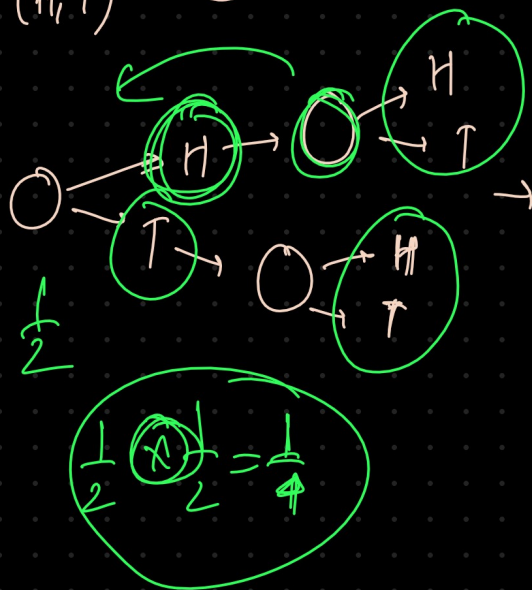
$$p(A) * p(B) = \frac{1}{13} * \frac{1}{13} = \frac{1}{221}$$

Toss 2 time

(H) (T)

p.s. Event $\rightarrow$ Head

$$P_c = 0 \rightarrow \frac{(H)}{(H, T)} = \frac{1}{2} = 0.5$$



AND $\rightarrow \times$

OR $\rightarrow +$

P. Drawing 2 cards (without replacement)

Event: Both Aces A or A

$$p(\text{first card}) \rightarrow \frac{4}{52}$$

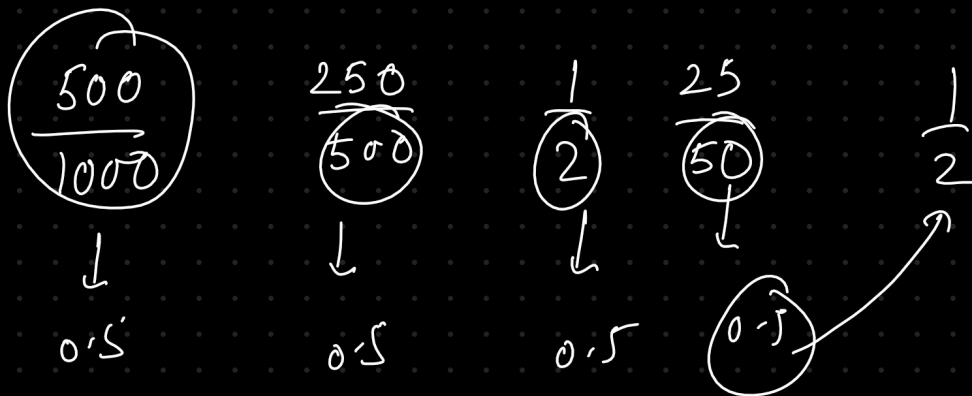
$$\frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$$

$$p(\text{second card}) \rightarrow \frac{3}{51}$$

P. Toss 3 coin

$$\text{Event} = H \geq 2$$

$$p(H \geq 2) = \frac{4}{8} = \frac{1}{2} = 0.5$$



$$\frac{1000}{2000} \leftarrow$$

Note:

$$P \rightarrow H \rightarrow \frac{1}{2} \rightarrow 0.5 \rightarrow$$

$$1, 2, 3, \quad \xrightarrow{\quad\quad\quad} 1000$$

←

$$H \rightarrow 0.48, \quad \frac{1}{2}$$

$$\frac{1}{2}$$

$$6 \rightarrow \boxed{6} \rightarrow \frac{1}{6}$$

$$H \rightarrow \uparrow$$

[Ace, Ace, Ace, Ace, Other, Other]

$\downarrow$
 4

$\downarrow$
 52

p.1 Bag with Balls

Bag $\rightarrow$ 5 red
 $\rightarrow$ 3 blue

$P(\text{Red}) \rightarrow \frac{5}{8}$
 $P(\text{Blue}) \rightarrow \frac{3}{8}$

$\rightarrow$ get 1 ball

p.1 Two Balls Drawn without replacement

Bag: 5 red, 3 blue

Event: Both red

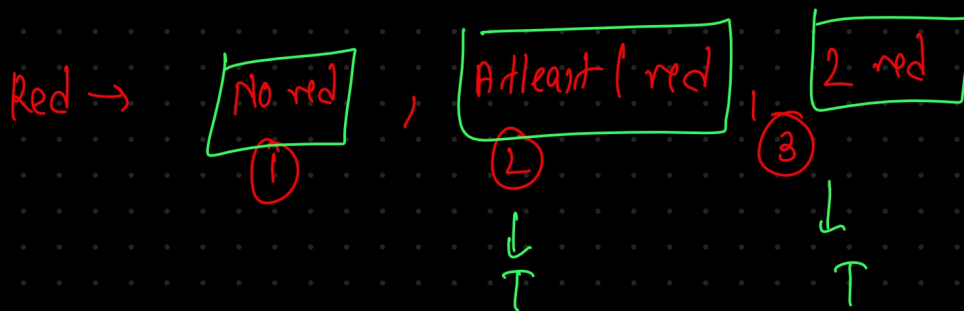
And | OR

$$\begin{aligned}
 P(\text{Both red}) &= P(\text{first-red}) \text{ AND } P(\text{second-red}) \\
 &= \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \underline{\underline{0.357}}
 \end{aligned}$$

with replacment:

$$= \frac{5}{8} \times \frac{5}{8}$$

p.): Atleast One red in two draws
 Bag: 5 red, 3 Blue (without replacement)



$$p(\text{Atleast 1 red}) = 1 - p(\text{No red})$$

$\downarrow$
 $p(\text{both blue})$

$$\frac{\textcircled{3}}{\textcircled{8}} \times \frac{2}{7} = \frac{6}{56} = \textcircled{0.107}$$

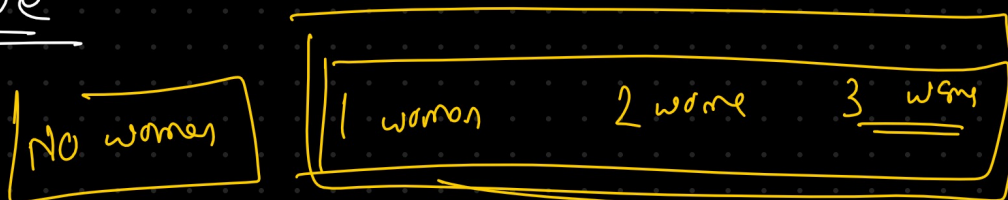
$$1 - 0.107 \rightarrow 0.893 \Rightarrow p(\geq 1 \text{ red})$$

p.5: selecting committee
 10 people $\rightarrow$ 6 men
 $\rightarrow$ 4 women

Event: Atleast 1 woman

select 3 people

0.83...



$\downarrow$
1

$$\text{No women} = 1 - (\geq 1 \text{ women})$$

$$\text{No. women} = \underline{\underline{3 \text{ men}}}$$

$$\frac{{}^6C_3}{{}^{10}C_3} = \frac{20}{120} = 0.166$$

$$P(\geq 1 \text{ woman}) = 1 - \text{No. ways}$$

$$1 - 0.166 = \underline{\underline{0.833}}$$

Conditional Probability

$P(A) \rightarrow$ Traffic Jam

$P(B) \rightarrow$ Rainy day

$$P(A) = 10\%$$

$$P(A|B) = \uparrow \rightarrow \underline{\underline{\text{dependent}}}$$

$$P(A|B) \neq P(A)$$

$P(A) \rightarrow$ coin flip is Head

$P(B) \rightarrow$ coin flip is Tail

$P(A) \rightarrow$ 0.5 or 50%

$P(A|B) \rightarrow$ Independent Event

$$P(A|B) = P(A)$$

Joint probability:

prob. of A & B happening.

→ dependent

→ independent

→ $A \cap B$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

or

$$P(A \cap B) = P(B|A) \cdot P(A)$$

Ex. 2 coin
→ Both Head

$$P(A) \rightarrow \frac{1}{2}$$

$$P(B|A) \rightarrow \frac{1}{2}$$

$$P(B) \rightarrow \frac{1}{2}$$

$$\left. \begin{array}{l} P(A) \cdot P(B|A) \\ P(A) \cdot P(B) \end{array} \right\}$$

depend : —

$$P(A \cap B) \rightarrow$$

$$P(A|B) \cdot P(B)$$

$$P(B|A) \cdot P(A)$$

independent : —

$$P(A \cap B) \rightarrow$$

$$P(A) \cdot P(B)$$

$$P(A \cap B) \rightarrow P(A|B) \cdot P(B)$$

$$\rightarrow P(A|B)$$

$$\downarrow$$

$$P(A) \cdot P(B)$$

2 con

1 → H

1 → H

(1) conditional prob.

→ prob of an event given another event has occurred.

dependent → without replace (pack of cards) →

independent → (coin flips)

traffic
↑
rain

stock price
↑
twitter

(2) Bayes Theorem

→ updates the probability of an event based on new information.

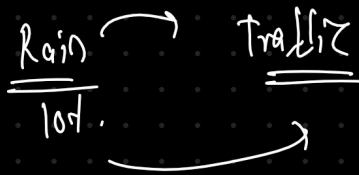
$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

C.P $\rightarrow$ what's the chance of A given B?

B.T $\rightarrow$ Given B, how does it change the chance of A?

$P(A) \rightarrow$ Traffic

$P(B) \rightarrow$ Rain
 $\rightarrow$ 

B.T $\rightarrow$ 

C.P $\rightarrow P(\text{Test} | \text{Disease})$

P. of Test given Disease

B.T $\rightarrow P(\text{Disease} | \text{Test})$

P. of D given test

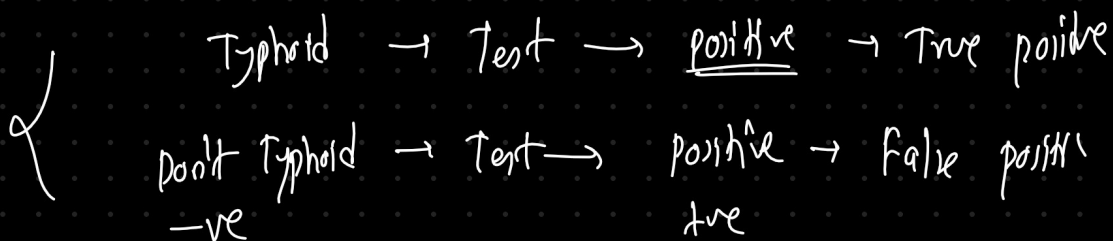
$P(D) \rightarrow 1\%$

$P(\text{pos} | D) = 99$

$P(D | \text{pos}) = ?$



Blood $\rightarrow D'$
 $\rightarrow$ Machine $\rightarrow 1\%$ (False positive)



Find $P(D | \text{pos}) = \frac{P(\text{pos} | D) \cdot P(D)}{P(\text{pos})}$

$\downarrow$ 0.01 $\downarrow$ 99%

$$P(\text{pos}) = P(\text{pos} | D) \cdot P(D) + P(\text{pos} | D') \cdot P(D')$$

$$P(\text{pos} | D) \cdot P(D)$$

$$P(\text{pos} | D) \cdot P(D) + P(\text{pos} | D') \cdot P(D') \rightarrow (0.99 \times 0.01) + (0.01 \times 0.99) \\ \rightarrow 0.0198$$

$$\frac{P(\text{pos} | D) \cdot P(D)}{0.0198} \rightarrow \frac{0.99 \times 0.01}{0.0198} = 0.5$$

There is only 50% chance you have the disease even after testing positive.

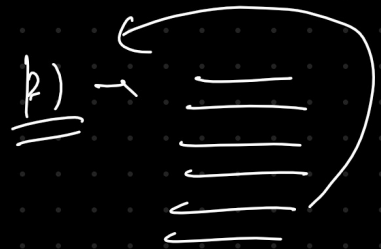
$$P(D | \text{pos}) \rightarrow \underline{P \text{ of } D \text{ given pos}}$$

$$P(\text{pos}) \rightarrow \begin{array}{l} D \rightarrow P(\text{pos} | D) \cdot P(D) \\ + \\ D' \rightarrow P(\text{pos} | D') \cdot P(D') \end{array}$$

p.) → write a really good p.)

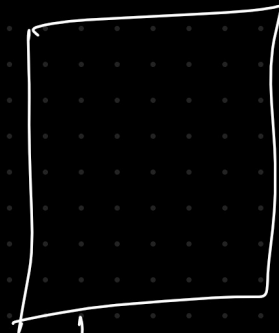
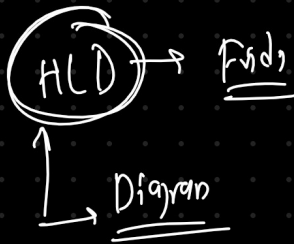
→ DB
→ python

p.)
Tool →
→ functionally →



Comm → p.) :

why :
Tool: visual

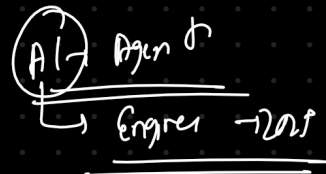


↓ starting a
project

SD

HLD of a project of your library

2025 



docter → AI + UI → profc

HLD →